

Second Certainty: Tax Management System
Financial Software Engineering Assignment (ECO5040S)

Cesaire Tobias (TBSCES001)

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1 Project Description

Second Certainty addresses one of life’s two certainties - **taxes** - by providing South African individuals and small businesses with an intuitive platform for managing tax liabilities throughout the fiscal year.

Traditional tax management often occurs reactively at tax submission deadlines, leading to unexpected liabilities, missed deductions, and stress. Second Certainty transforms this approach by implementing a proactive, year-round tax management system that calculates estimated tax liabilities based on real-time income and expense streams.

The system combines up-to-date South African tax regulations with personal financial data to deliver accurate tax calculations, visualizations, and insights, helping users optimize their tax position throughout the year.

2 Technical Requirements

2.1 Backend

- **Framework:** FastAPI 0.110.1 - intuitive api framework with modern Python type annotations
- **ORM:** SQLAlchemy 2.0.40 - provides flexible database operations and ORM capabilities
- **Validation:** Pydantic 2.6.0 with pydantic-settings 2.2.1 - ensures robust data validation with minimal code
- **Database:**
 - PostgreSQL (production) - chosen for native support in Render (for deployment)
 - SQLite (development)
- **Authentication:** JWT with HTTPBearer security and bcrypt password hashing - secure authentication approach
- **Testing:** Pytest 7.4.4 with pytest-asyncio 0.23.5 - testing framework for Python including async support
- **Documentation:** OpenAPI (Swagger UI) - automatic API documentation generation
- **Web Scraping:** BeautifulSoup4 4.12.3 with lxml 5.1.0 - for parsing SARS tax data
- **Database Migrations:** Alembic 1.13.1 - database schema management
- **Server:** Uvicorn 0.29.0 - ASGI server

2.2 Frontend

- **Framework:** React 18.2 - project specified
- **Routing:** React Router 6.22.1 - provides declarative routing for React applications

- **HTTP Client:** Axios 1.6.7 - HTTP client for API communication
- **State Management:** React Context API - built-in state management solution that avoids external dependencies
- **Styling:** Tailwind CSS 3.4.1 - CSS framework for UI development
- **Data Visualization:** Recharts 2.12.0 - responsive charting library for React
- **Build Tools:** Create React App - simplified configuration and build process

2.3 Development Tools

- **Code Formatting:** Black 24.2.0, isort 5.13.2 - automated code formatting and import sorting
- **Linting:** Flake8 7.0.0 - code quality and style checking
- **Environment Management:** python-dotenv 1.0.1 - secure environment variable management

3 Application Features

3.1 Core User Management

- **Secure Authentication:** JWT-based authentication system with HTTPBearer tokens and bcrypt password hashing
- **User Registration and Login:** User account management with profile customization
- **Provisional Taxpayer Support:** Special handling for users subject to provisional tax requirements
- **Admin Features:** Administrative capabilities for system management and tax data updates

3.2 Financial Data Management

- **Multi-Source Income Tracking:** Support for salary, freelance, investment, rental, and other income types
- **Comprehensive Expense Management:** Categorized tax-deductible expenses with predefined types and validation
- **Real-time Calculations:** Immediate tax liability updates as financial data changes
- **Historical Data:** Multi-year tax data management with automatic tax year detection

3.3 Tax Calculations

- **South African Tax Compliance:** Current tax brackets, rebates, and thresholds for all age groups
- **Provisional Tax Management:** Automated calculation and tracking of provisional tax payments for eligible taxpayers
- **Age-based Calculations:** Automatic rebate and threshold adjustments based on user age

3.4 Data Integration and Management

- **SARS Website Integration:** Automated scraping of latest tax rates with multiple fallback strategies

- **Robust Data Pipeline:** Multi-tiered approach for tax data retrieval including manual fallbacks
- **Database Migration System:** Alembic-powered schema management for seamless updates
- **Comprehensive Logging:** Structured logging with file rotation and error tracking

3.5 Security and Performance

- **Input Validation:** Data validation using Pydantic schemas
- **CORS Management:** Secure cross-origin resource sharing configuration
- **Error Handling:** Error responses and exception management

4 Deployment

The Second Certainty application is deployed on Render’s free tier, which may result in slow initial load times after periods of inactivity. When logging in, please note that you may need to retry after **No response from server** message - this is a function of deployment on Render’s free tier which experiences “cold starts” after 15 minutes of inactivity.

Live Application URLs:

- **Frontend Application:** <https://second-certainty.onrender.com>
- **Backend API:** <https://second-certainty-api.onrender.com>
- **API Documentation:** <https://second-certainty-api.onrender.com/api/docs>

Performance Considerations:

- Initial load times may exceed 30 seconds due to service “spin-up” on free tier
- Subsequent requests are responsive once the service is active
- Database queries are optimized for performance despite platform limitations

For more detailed information, please refer to the following documentation:

- **Installation Guide:** <https://github.com/ces0491/second-certainty/blob/main/docs/INSTALLATION.md>
- Complete setup instructions for local development
- **Deployment Guide:** <https://github.com/ces0491/second-certainty/blob/main/docs/DEPLOYMENT.md>
- Comprehensive production deployment instructions
- **API Documentation:** <https://github.com/ces0491/second-certainty/blob/main/docs/API.md>
- Detailed API reference with examples

5 Code Repository

The project is maintained in two separate repositories:

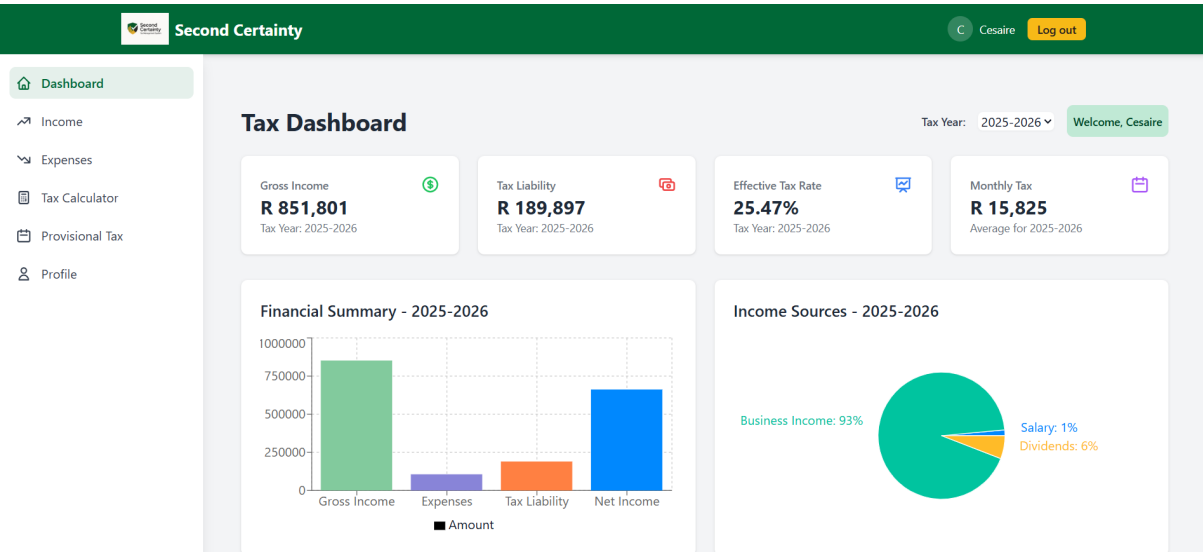
- **Backend Repository:** <https://github.com/ces0491/second-certainty>
- **Frontend Repository:** <https://github.com/ces0491/second-certainty-frontend>

The repositories include comprehensive documentation covering:

- **Installation and Setup:** Step-by-step local development environment setup
- **API Documentation:** Complete endpoint reference with request/response examples
- **Database Schema:** Detailed model relationships and migration procedures
- **Testing Framework:** Unit, integration, and API testing suites
- **Deployment Procedures:** Production deployment with security best practices

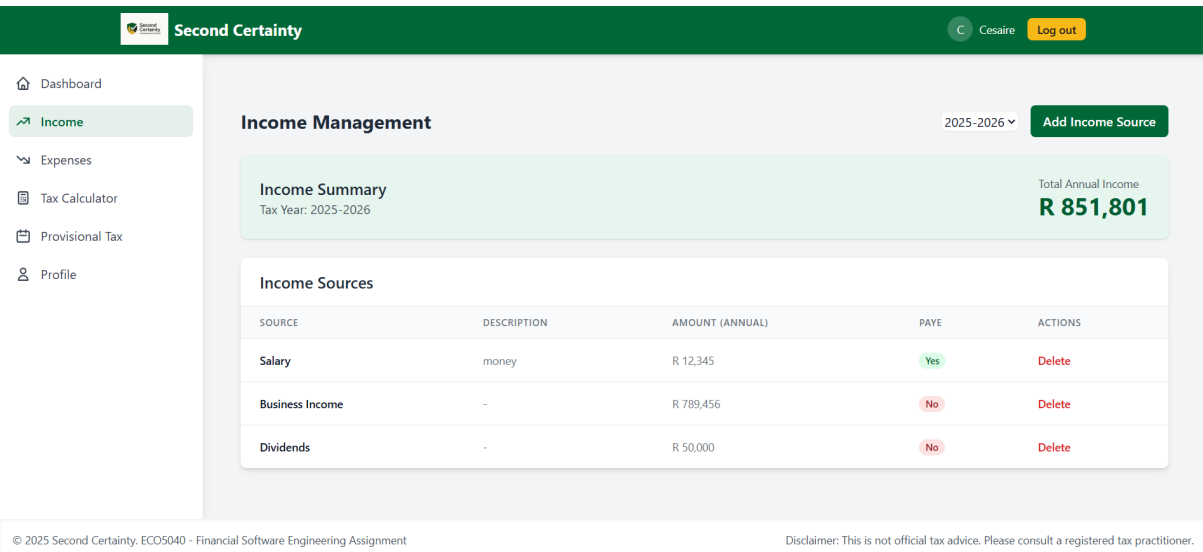
6 Screenshots

6.1 Dashboard



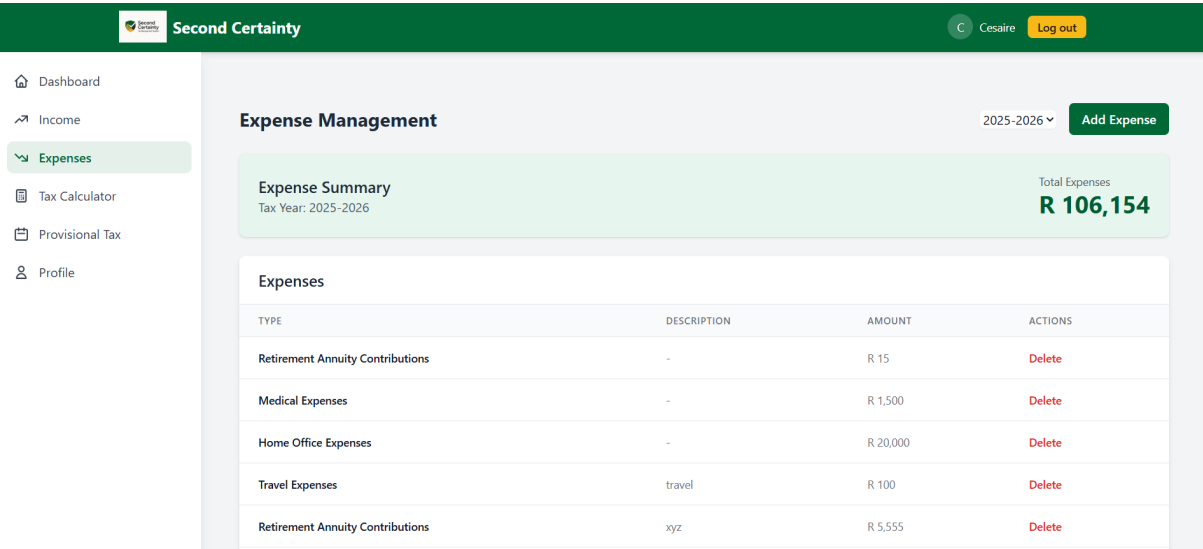
The main dashboard showing key tax metrics and income/expense breakdown.

6.2 Income Tracking



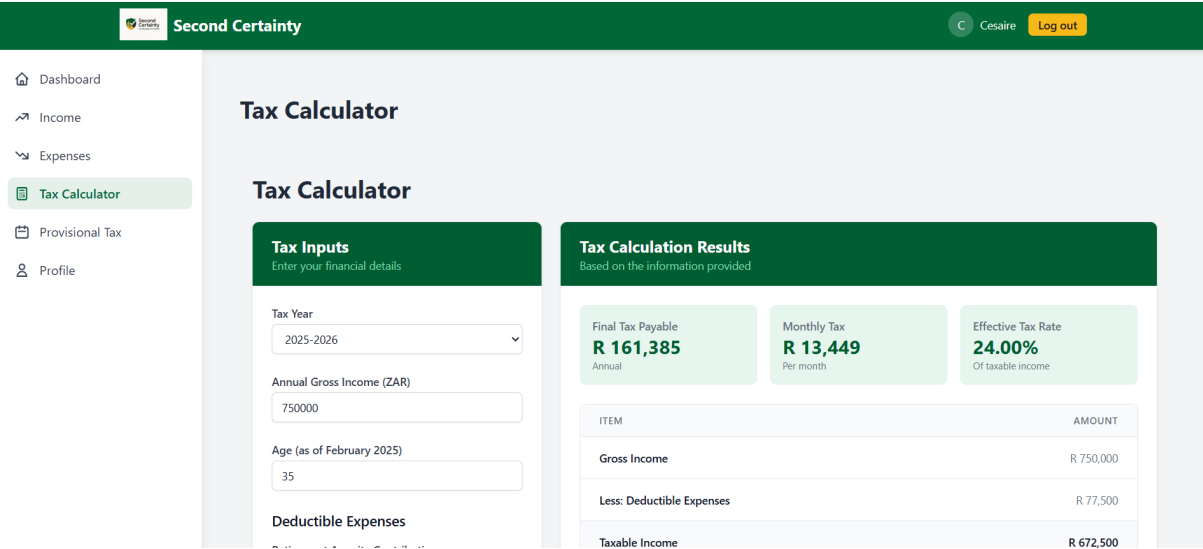
Interface for managing multiple income sources with PAYE categorization and annual amount tracking.

6.3 Expense Management



Expense tracking interface with predefined tax-deductible categories.

6.4 Tax Calculator



Interactive tax calculator showing real-time liability calculations based on current financial data with age-based rebate adjustments.

6.5 Provisional Tax Planner

The screenshot shows the 'Provisional Tax' section of the Second Certainty application. The interface includes a sidebar with navigation links: Dashboard, Income, Expenses, Tax Calculator, Provisional Tax (selected), and Profile. The main content area is titled 'Provisional Tax' and shows the fiscal year '2025-2026'. A 'Refresh Tax Calculation' button is present. Below this, there are two columns for 'First Provisional Payment' and 'Second Provisional Payment'. Both show a 'Due Date' of 'Not available' and an 'Amount' of 'R 0'. The 'Payment Details' for the first payment state it represents 50% of the estimated annual tax liability, while the second represents the remaining 50%. At the bottom, a 'Provisional Tax Summary' table shows 'Total Annual Tax' as 'R 0', 'Taxable Income' as 'R 0', and 'Effective Tax Rate' as '0.00%'. A yellow progress bar is at the very bottom.

Provisional Tax Summary		
Total Annual Tax	Taxable Income	Effective Tax Rate
R 0	R 0	0.00%

Tool for calculating and tracking provisional tax payments throughout the fiscal year.

6.6 Profile Management

The screenshot shows the 'User Profile' section of the Second Certainty application. The sidebar is the same as in the previous screenshot, with 'Profile' selected. The main content area is titled 'User Profile' and contains two sections: 'Profile Information' and 'Account Security'. The 'Profile Information' section has an 'Edit Profile' button and displays the following details: First Name 'Cesaire', Last Name 'Tobias', Email Address 'cesairetobias@gmail.com', Date of Birth '2025/05/14', and Provisional Taxpayer Status 'Yes'. The 'Account Security' section has a 'Change Password' button and a note: 'For security reasons, please use a strong password that you don't use elsewhere.'

User profile interface for managing personal information, provisional taxpayer status, and account preferences.

7 Technical Architecture

7.1 Backend Architecture

- **RESTful API Design:** Consistent endpoint structure following REST principles
- **Modular Code Organization:** Separation of concerns with dedicated modules for authentication, tax calculations, and data management
- **Async Processing:** Non-blocking operations for web scraping and external API calls
- **Database Abstraction:** SQLAlchemy ORM with migration support for schema evolution

- **Error Handling:** Exception management with structured error responses

7.2 Frontend Architecture

- **Component-Based Design:** Reusable React components with clear separation of concerns
- **State Management:** Context API for global state with local component state for UI interactions
- **Responsive Design:** Tailwind CSS implementation for all screen sizes
- **Data Visualization:** Interactive charts and graphs using Recharts library
- **API Integration:** Centralized Axios configuration with error handling and retry logic

7.3 Security Implementation

- **Authentication Flow:** JWT tokens with HTTPBearer security and automatic token refresh
- **Password Security:** Bcrypt hashing with salt for secure password storage
- **Input Sanitization:** validation at both API and database levels
- **CORS Configuration:** Secure cross-origin resource sharing with whitelisted domains

8 Future Enhancements

- **Rate Limiting:** Implement rate limiting
- **Profile Deletion:** Enable safe deletion of profiles
- **Improve Standardization:** For logging, error handling, user input sanitization
- **Enhanced Visualizations:** Use more sophisticated charting libraries like Highchart for better visualizations
- **Mobile Application:** React Native app for iOS and Android with offline capability
- **Advanced Reporting:** PDF generation for tax reports and provisional tax statements
- **Email Notifications:** Automated reminders for provisional tax due dates and tax season
- **Enhanced Security:** Two-factor authentication and advanced session management
- **Banking Integration:** Direct API connections with South African banks for automatic transaction import
- **Advanced Analytics:** Machine learning-powered tax optimization recommendations
- **Multi-User Support:** Family and business account management with role-based permissions
- **Tax Practitioner Features:** Professional tools for accountants and tax consultants